

Lab: Derivation Exercise — Generative Models and Variational Inference

Objective

Derive the variational inference update equations for a concrete generative model, connecting the abstract mathematics to explicit computations.

Part 1: Hidden Markov Model Specification

Goal: Write out a complete generative model.

Consider a Hidden Markov Model with:

- $K = 3$ hidden states ($s \in \{1, 2, 3\}$)
- Observation model: $p(o_t | s_t = k) = N(o_t; \mu_k, \sigma_k^2)$ with known parameters
- Transition model: $p(s_t | s_{t-1})$ given by a 3×3 matrix $\mathbf{A}$ with rows summing to 1
- Prior: $p(s_1) = \text{categorical}$ with probabilities $[0.5, 0.3, 0.2]$

Task: Write the complete joint distribution $p(o_{1:T}, s_{1:T})$ for $T = 3$ timesteps. Expand the product notation fully.

Write your response here

Part 2: Variational Free Energy for the HMM

Goal: Derive the free energy for this model.

Let $q(s_{1:T}) = \prod_{t=1}^T q(s_t)$ be the mean-field recognition density, where each $q(s_t)$ is a categorical distribution with parameters (probabilities) $r_t = [r_{t1}, r_{t2}, r_{t3}]$.

Task: Write the variational free energy:

$$F = E_q[\ln q(s_{1:T})] - E_q[\ln p(o_{1:T}, s_{1:T})]$$

Expand each term. The first term is the negative entropy of q . The second term involves the log-likelihood and log-transition probabilities.

Write your response here

Part 3: Optimal Update Equations

Goal: Derive the fixed-point equations for the sufficient statistics.

Taking the derivative $\partial F / \partial r_{ij} = 0$ (with the constraint $\sum_j r_{ij} = 1$), derive the optimal $q^*(s_i)$:

$$\ln q^*(s_i = k) \propto \ln p(o_i | s_i = k) + E_{-}\{q(s_{i-1})\} [\ln p(s_i = k | s_{i-1})] + E_{-}\{q(s_{i+1})\} [\ln p(s_{i+1} | s_i = k)]$$

Show all steps. Explain why the update depends on messages from both the past (s_{i-1}) and the future (s_{i+1}).

Write your response here

Part 4: Gaussian State-Space Model

Goal: Repeat the derivation for a continuous model.

Consider a linear Gaussian state-space model:

- $p(s_t | s_{t-1}) = N(s_t; A \cdot s_{t-1}, Q)$ — state dynamics
- $p(o_t | s_t) = N(o_t; C \cdot s_t, R)$ — observation model
- $p(s_1) = N(s_1; \mu_0, \Sigma_0)$ — prior

With recognition density $q(s_t) = N(s_t; m_t, V_t)$.

Task: Derive the update equations for the sufficient statistics m_t (mean) and V_t (covariance) by minimizing F. Show that these correspond to Kalman filter/smoothing equations.

Write your response here

Part 5: Synthesis

In 200 words, explain the common mathematical structure across Parts 2-4: in both discrete and continuous cases, the optimal recognition density arises from minimizing free energy, and the update equations combine likelihood (data-driven) terms with transition (model-driven) terms.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Model specification	Writing the HMM joint distribution
2	Free energy derivation	Expanding F for mean-field HMM
3	Variational calculus	Deriving optimal update equations
4	Continuous extension	Gaussian state-space / Kalman filtering
5	Conceptual synthesis	Unifying discrete and continuous cases